

Reference excerpt 01-013

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jugular-heart segment. The z axis is positive in the jugular-heart direction. l is the distance between the RA and point G in which the scan takes place. Blood velocity is indicated by w . propagation velocity c . Such pressure waves are responsible for the modulated component of the velocity of blood in the IJV. For this reason, their wave equation can be used to derive the instantaneous velocity of the blood even in the absence of a direct measurement of such parameter obtained, for example, by using an ultrasound Doppler scanner. A first attempt to this goal was presented in [Sisini et al.(2015b)], where the instantaneous velocity of the blood in the IJV was determined qualitatively, for one cardiac cycle, using the Womersley equation [Womersley(1955a), Womersley(1955b), Womersley(1955c)].

Physical description of the RA-IJV segment

The RA-IJV system is represented in Fig. 1 as a tube with a circular section. G indicates a reference point on the right IJV. The $w(z, r, t)$ function represents the velocity of the blood in the z direction. It depends on z coordinate, on r (the distance from the axis z) and on t (the time). The symbol $w(z, t)$ is used to indicate the blood velocity averaged over the CSA. The pressure at the right end of the tube ($z = 0$) is the residual arterial pressure and it is assumed to have a constant value p_r while at the left end of the tube ($z = z_{RA}$) the pressure feels the effects of the RA activity